Control a LoRa iGate from Graywolf

Illustrated walkthrough • Revised 24 October 2026 • Graywolf RXT fork

The principle: install the same key on the iGate and in Graywolf, then use the ⚡ panel buttons to change settings.

This walkthrough requires RXT iGate firmware with Remote Command Authentication and a Graywolf RXT build offering CA2RXU authenticated control.

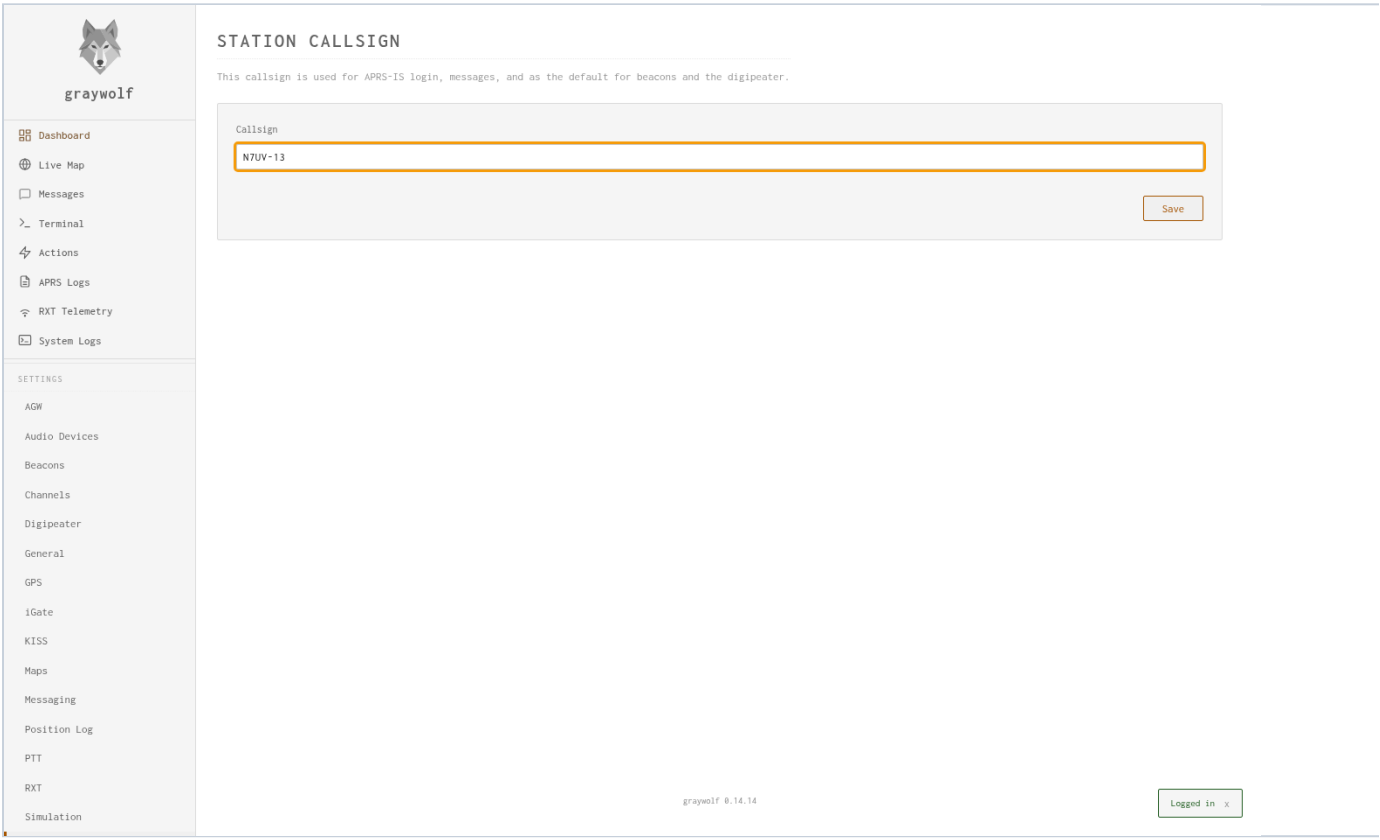
1. Identify the sending station

In Graywolf, open **Settings** → **Station Callsign**. Enter **N7UV-13**, then click **Save**.

This is the controller's identity, including its SSID. It must exactly match the controller authorized on the iGate.

Controller: Graywolf	Target station: iGate
N7UV-13	**W9EN-44**

These callsigns are examples. This walkthrough works when the RF address and the iGate's **Callsign - SSID** field are identical, here **W9EN-44**. If the device uses a different tactical callsign, address its on-air name and enter its Station Callsign-SSID as the signed identity, as described below.



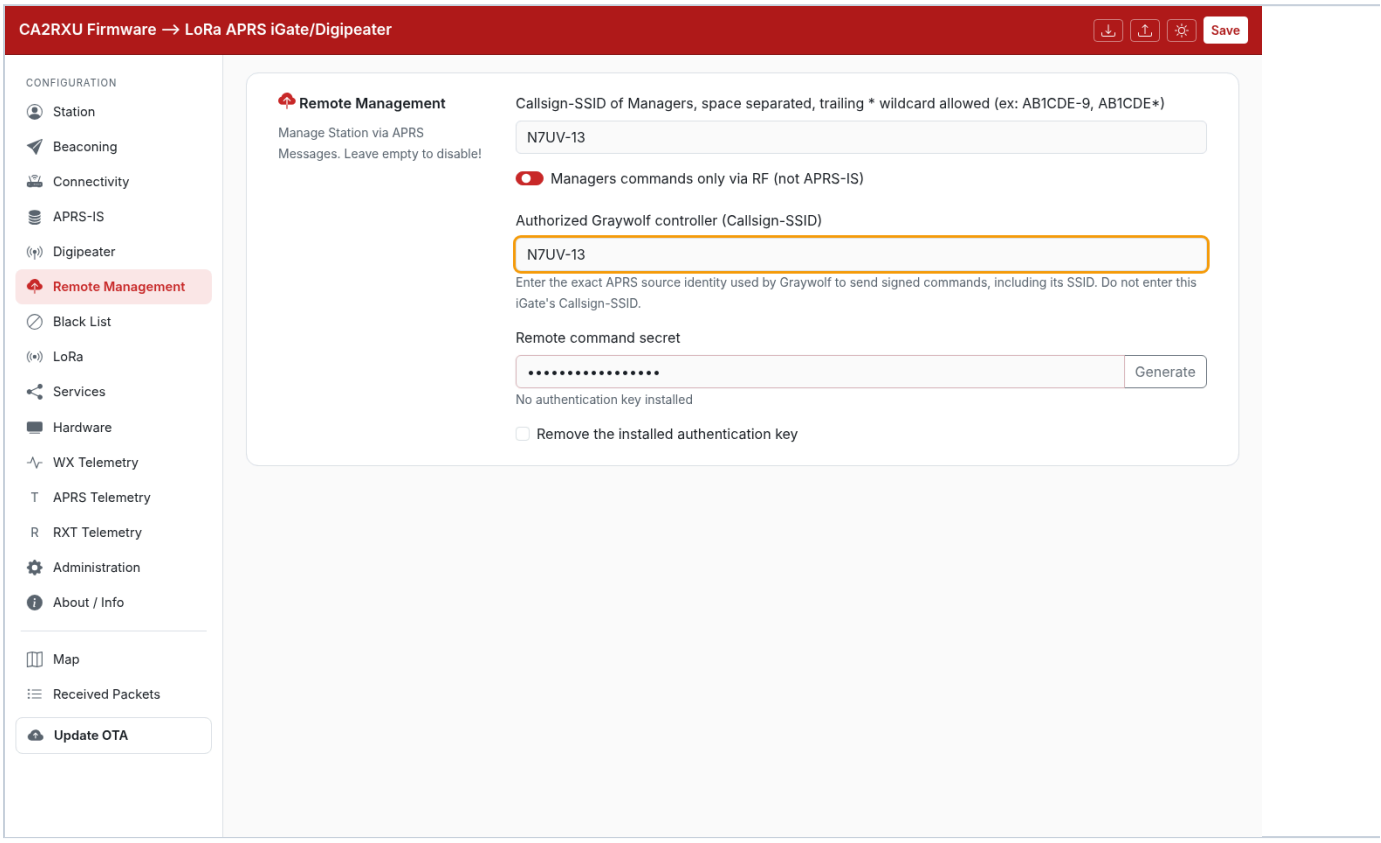
Graywolf: the Callsign field identifies the station sending commands.

2. Authorize Graywolf on the iGate

In the WebUI of iGate **W9EN-44**, open **Remote Management**.

1. Under **Callsign-SSID of Managers**, add **N7UV-13**, keeping any existing managers. This field allows status queries, among other functions.
2. Under **Authorized Graywolf controller (Callsign-SSID)**, enter **N7UV-13**. Do not enter W9EN-44: this field identifies the station sending commands.
3. For this radio-based walkthrough, enable **Managers commands only via RF (not APRS-IS)**.
4. For a first installation, click **Generate** next to **Remote command secret**. Copy the complete **43-character** key and keep it for step 5.
5. Click **Save** at the top of the page.

> **Already have a key installed?** Reuse the matching copy. An empty field does not mean the key is missing: the iGate does not display its saved key again. Do not generate a new key on only one device.



The firmware's Remote Management screen, rendered locally with demonstration values. No deployed station was reconfigured.

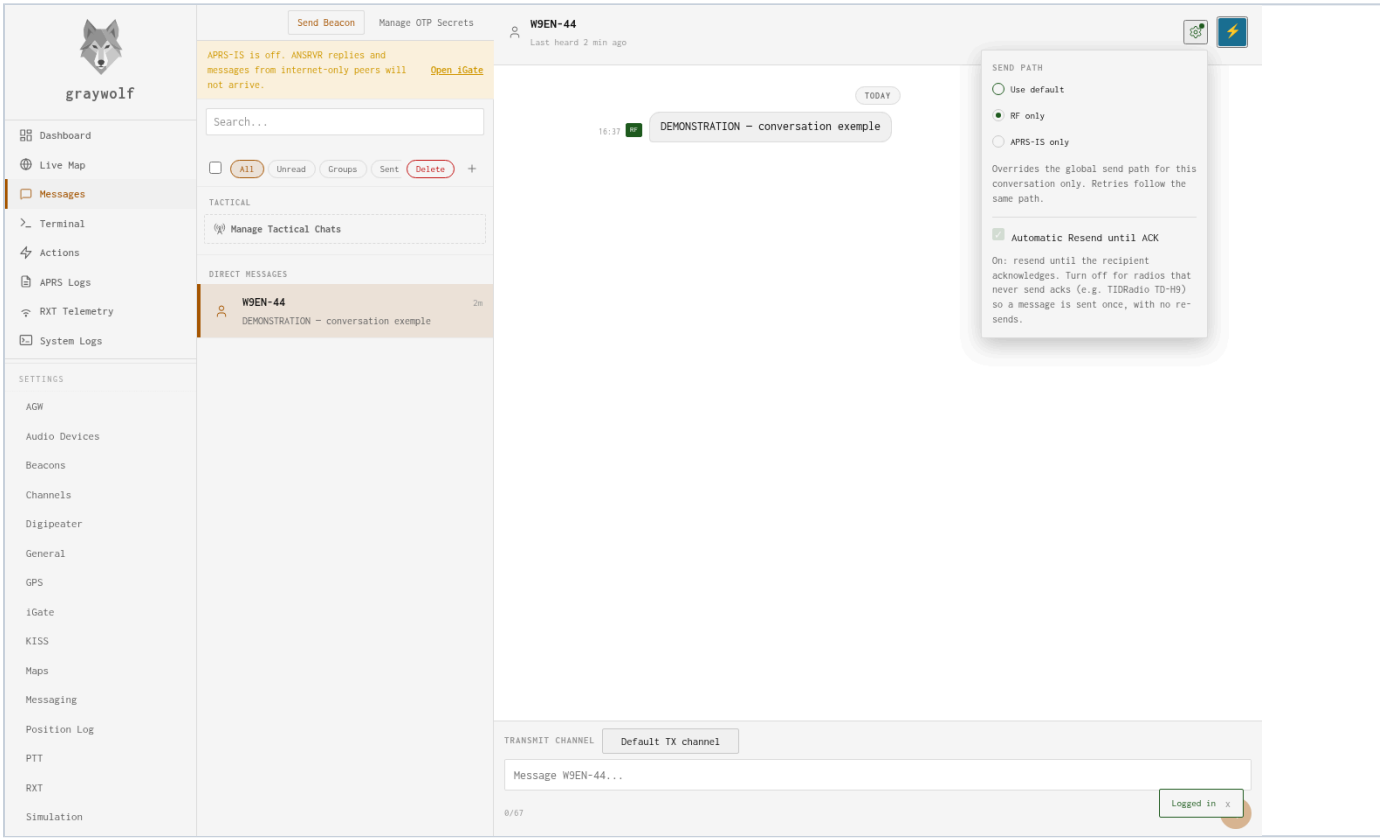
3. Choose the destination and RF route

In Graywolf, open **Messages**, then the direct conversation with **W9EN-44**.

If you do not have a conversation yet: first open **Settings** → **Messaging**, select the **RF only** send policy and your LoRa TNC channel. Then click **New message**, enter **W9EN-44** in **To**, and send `?APRSV` to start the conversation.

1. In the conversation header, click the small **gear** button: **Conversation routing**.
2. Select **RF only**, then close the menu.
3. Below the conversation, under **Transmit channel**, select the channel connected to your LoRa TNC. Its name depends on your configuration; the screenshots use **LoRa TNC**.

> The TNC connected to Graywolf must allow transmission. A receive-only RXT/JSON feed cannot send commands. When using a TNC2 connection, its TX source identity must also match **N7UV-13**.



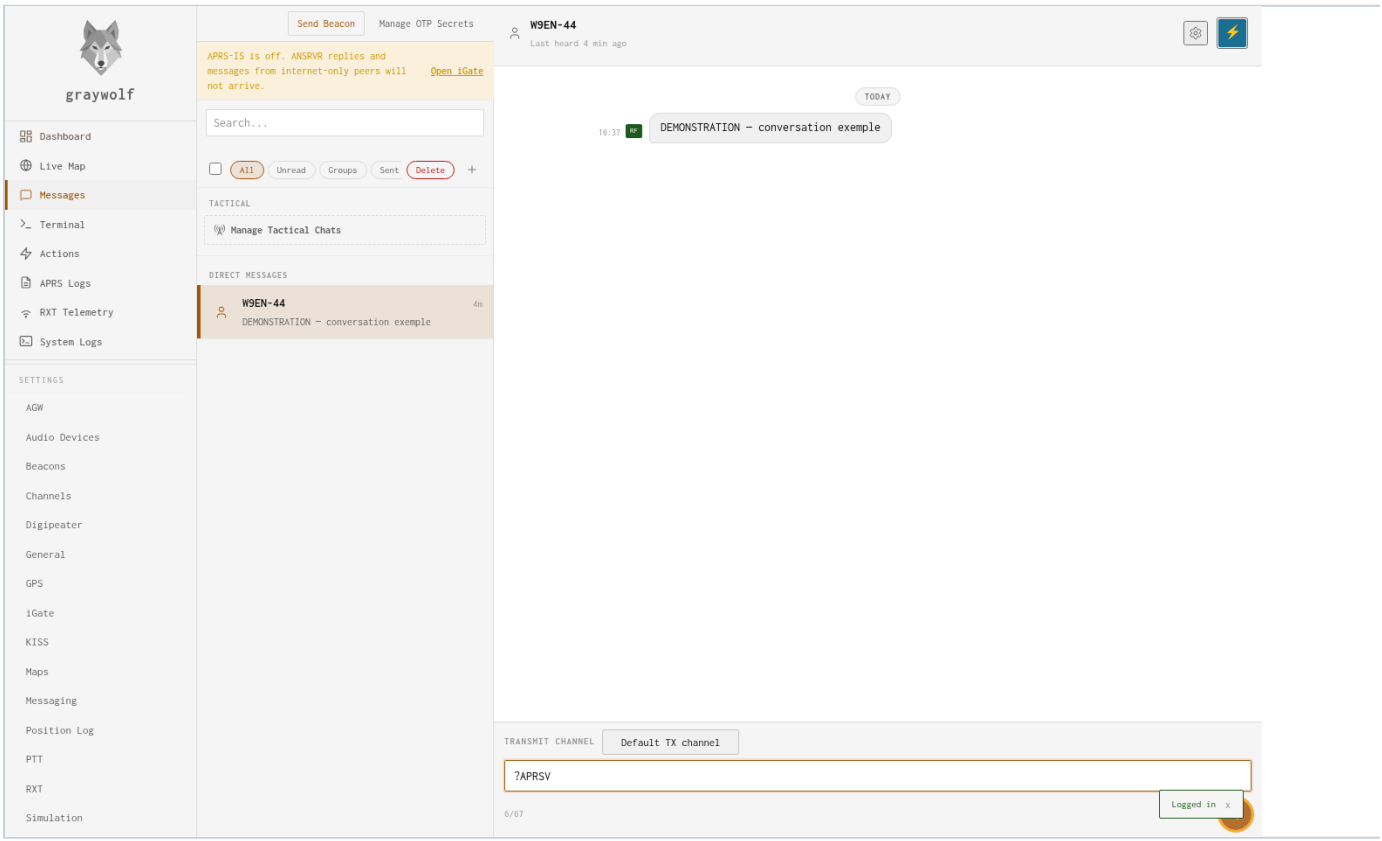
The routing menu sets this conversation's send path. Select the radio channel separately at the bottom.

4. Check a simple exchange first

1. In the normal message composer for **W9EN-44**, type `?APRSV`.
2. Click the **send arrow** at the bottom right.
3. Wait for the iGate's version reply, starting with `CA2RXU_LoRa_iGate v`.

This query changes no settings and needs no signature. It checks the message round trip before you test the key.

No reply? First check the destination, channel, TNC transmit permission and RF path. A missing reply does not yet show that the key is wrong.



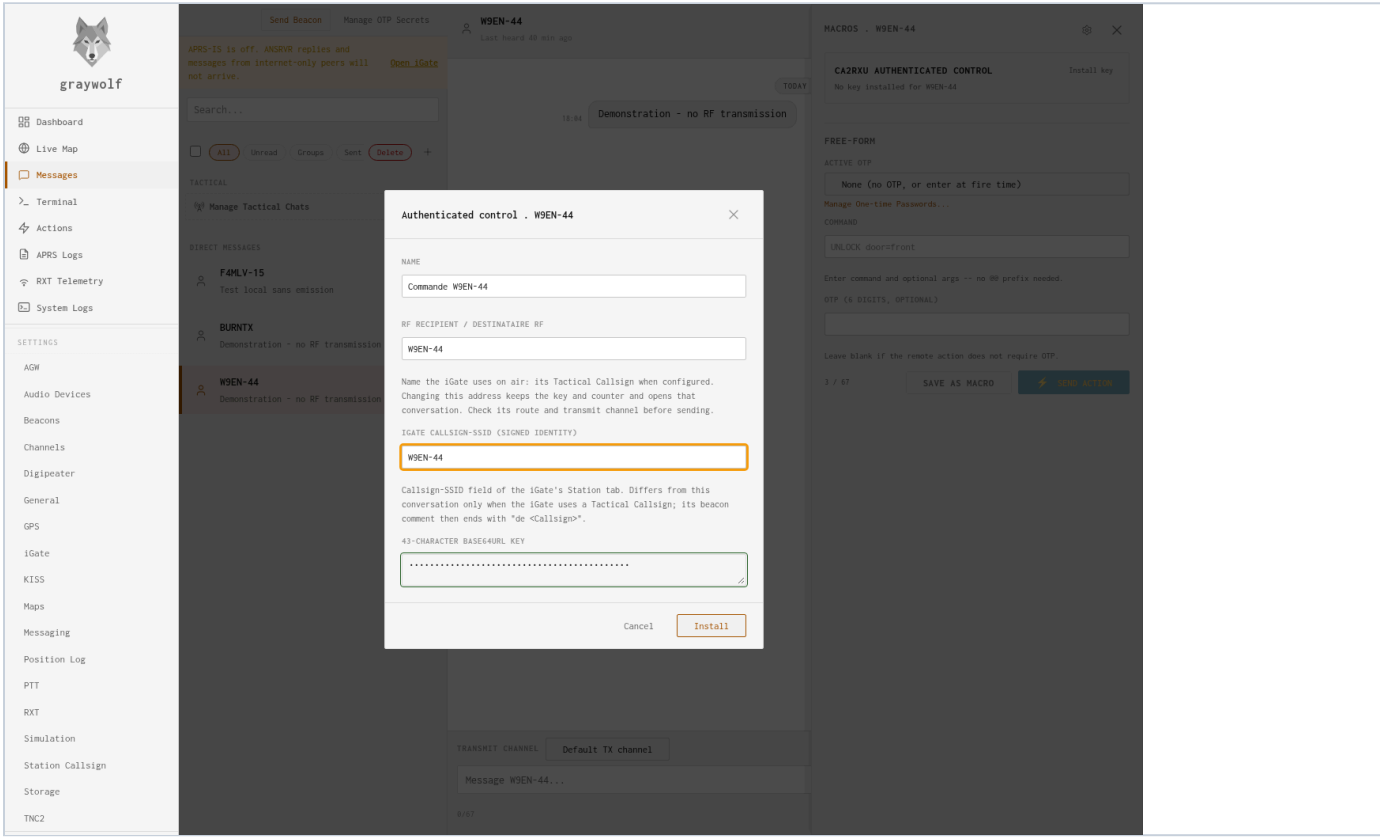
The query is prepared in the composer. It was not transmitted while taking these screenshots.

5. Install the same key in Graywolf

1. In the header of that same conversation, click the ⚡ button.
2. In the drawer on the right, find **CA2RXU authenticated control**, then click **Install key**.
3. Under **Name**, enter a descriptive name, such as **W9EN-44 control**. This label is up to you.
4. Under **iGate Callsign-SSID (signed identity)**, enter the iGate's **Callsign - SSID** from its Station tab, here **W9EN-44**.
5. Under **43-character Base64URL key**, paste **the key copied from the iGate in step 2**, then click **Install**.

This is neither your Graywolf login password nor your APRS-IS passcode. A separate credential is stored for each target iGate.

> The **Manage OTP Secrets** button and the macros in the same drawer belong to a different mechanism. For this walkthrough, use only **CA2RXU authenticated control**.



Installing the key for W9EN-44. The demonstration value is masked in the screenshot.

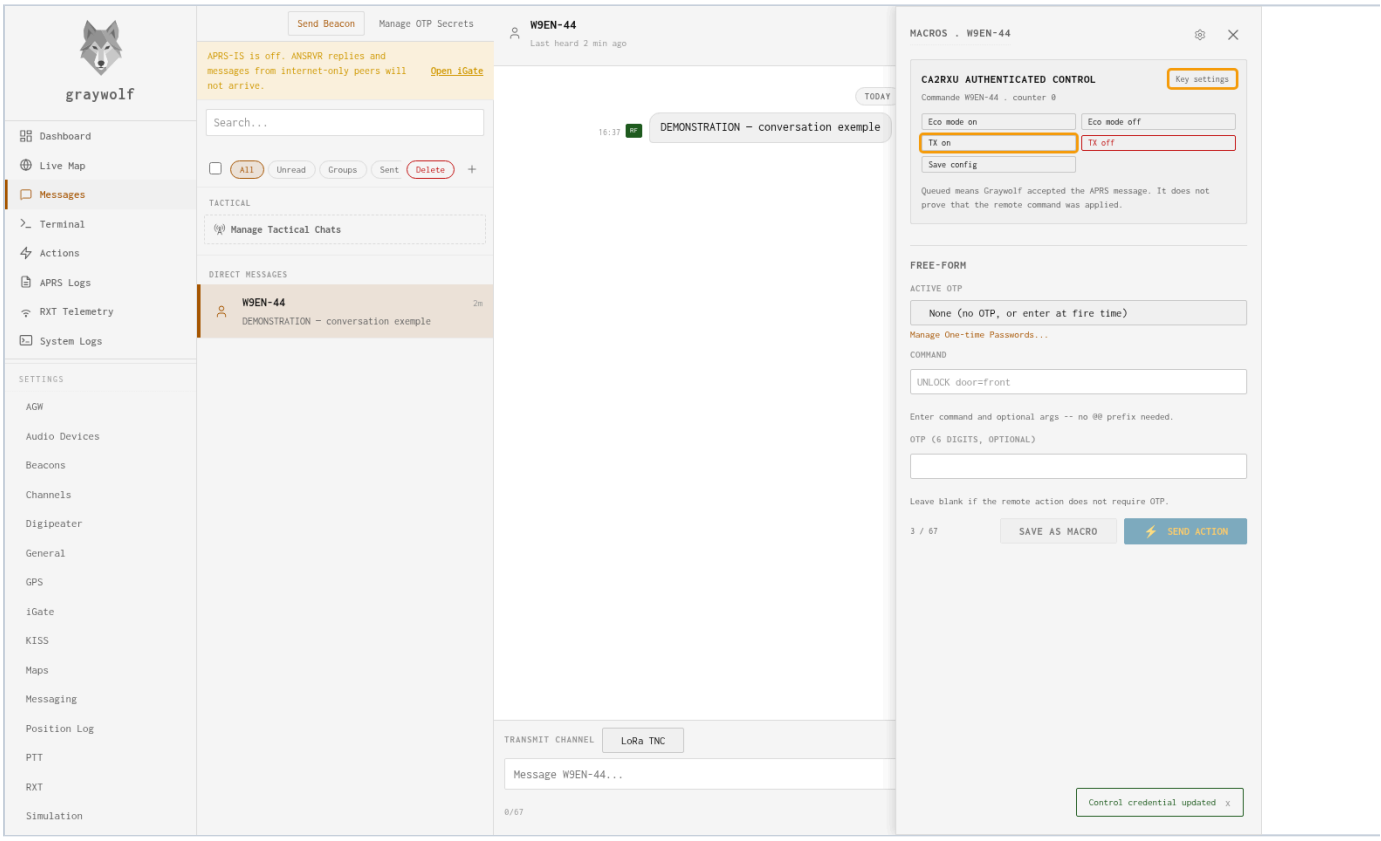
6. Send your first signed command

Once the key is installed, the **CA2RXU authenticated control** panel shows five buttons.

1. For the first test, click **TX on**. If the iGate is already transmitting, this simply confirms the enabled state.
2. Graywolf automatically prepares and signs the command. It displays **TX=ON queued**, followed by a counter.
3. Look for the iGate's reply in the conversation: `TX=ON` or `TX was ON`.
4. You can then type `?TX=?` in the normal composer to read back the state.

> **Queued means accepted for sending, not executed.** Check the reply and resulting state. An APRS ACK confirms message receipt, not authenticated execution of the setting change.

Do not type `?TX=ON` in the composer for this test: it would be an unsigned command, even with a key installed.



These buttons sign setting changes. No command button was activated while taking the screenshots.

Devices with a tactical callsign

A device can have **N7UV** in the Station tab's **Callsign - SSID** field and **BURNTX** as its **Tactical Callsign**. These values serve different purposes.

Setting	Example
Authorized Graywolf controller	N7UV-13
RF destination and Graywolf conversation	BURNTX
Target identity used to sign	N7UV — the device's Callsign - SSID field

The published firmware verifies the signature with **N7UV**, its Station Callsign-SSID. Graywolf addresses **BURNTX** while signing for **N7UV**.

> **Configure the signed identity:** open the **BURNTX** conversation, then ⚡ → **Key settings** (or **Install key**). Set **RF recipient / Destinataire RF** to **BURNTX** and **iGate Callsign-SSID (signed identity)** to **N7UV**. If the key was installed under N7UV, edit that existing key instead of installing it again. Saving opens the RF destination conversation; check its route and channel. Leave the secret blank when editing an existing key and click **Save**: the key and counter are preserved. No firmware update is needed.

Keep **BURNTX** as this device's RF command recipient. The device answers RF messages addressed to its tactical callsign. Ordinary messages and unsigned queries such as `?APRSV` also remain addressed to BURNTX.

Read the device's **Callsign - SSID** field to identify the value needed for signing. Its beacon's **de N7UV** suffix can also provide this information.

After the first test

What the buttons do

Button	Effect
Eco mode on / off	Enables or disables eco mode, when supported by the iGate's operating mode.
TX on / off	Enables or disables radio transmission. TX off asks for confirmation and may interrupt RF replies.
Save config	Saves the current configuration for the next startup. Use it after making the desired change, not to install the key in Graywolf.

Queries stay in the normal conversation

`?APRSV` reads the version. `?TX=?` reads TX state and `?EM=?` reads eco-mode state. The latter two queries require the sender to match the managers list and obey the RF-only restriction. They are not signed using the buttons.

If something does not work

What you see	What to check
No CA2RXU panel	Check that your Graywolf RXT build includes this feature, that this is a direct conversation, and that you clicked ⚡ in its header.
Install key, but no five buttons	No key is stored for the exact callsign of this conversation.
AUTH REQUIRED	An unsigned setting change was received. Use the corresponding button in the ⚡ panel.
AUTH FAILED	Check the matching key, controller N7UV-13 and signed target identity, which must match the iGate's Callsign - SSID field. With a tactical callsign, check the separate signed identity in Key settings while keeping the on-air name as recipient. The firmware console gives the precise rejection reason.
No reply	Repeat the ?APRSV check. Then check identities and transport restrictions: an unauthorized controller may be silently ignored.

> **Do not delete and reinstall the same key to "fix" the counter.** Graywolf would restart its counter while the iGate still remembers commands already received. If reprovisioning is necessary, install a fresh matching key on both devices.

Once setup is complete, later use is simply: **Messages → iGate conversation → ⚡ → command button**.

About the screenshots

Graywolf screenshots were taken on an isolated local instance, binary e6e8447f; documentation reference 4c4314b2. Callsigns and conversation data are demonstration examples. The iGate form was rendered from the RXT repository's WebUI files without connecting to a device. No operational secret is shown, and no RF transmission or APRS-IS connection was used for this walkthrough. The key-installation screenshot was updated on 24 October 2026 on an isolated instance with the signed-identity field; the remaining screenshots predate this field. No command was transmitted during the update. These illustrations are not an end-to-end radio test.

Detailed reference: Remote Actions — Send an authenticated iGate command. Version française.